

Revision Notes for Class 7 Maths Chapter 1 Geometric Twins

Geometry is about understanding shapes and their properties. When you see two figures that look exactly the same in both shape and size, those figures are called congruent. To check if two shapes are congruent, you can try placing one over the other, even after rotating or flipping it, and if they fit perfectly, they are congruent.

When recreating a symbol or drawing, you can't always rely just on tracing, especially if it is large. Instead, taking measurements such as the lengths of sides and the angles can help you make an exact copy. For example, knowing the lengths $AB = 4$ cm and $BC = 8$ cm of a symbol is not enough to recreate it. There can be multiple possibilities with those side lengths. But if you also know the angle between them ($\angle ABC = 80^\circ$), then only one unique figure can be drawn.

Congruence of Figures

When figures are congruent, they can be matched exactly—side to side, angle to angle. Even when you spin or reflect a figure, it can still be congruent. For basic shapes like circles and rectangles, you need to check their important measurements: the radius for circles (same radius means congruent circles) and the length and width for rectangles (both pairs must be equal for congruence).

- All congruent figures have the same size and shape.
- To confirm congruence, superimpose (place) one figure over another after rotating or flipping.
- A circle's radius and a rectangle's length and width must match for congruence.

Congruence of Triangles

A huge theme in geometry is the congruence of triangles. Two triangles are congruent if all three sides of one triangle are exactly equal to the three sides of the other. This is called the SSS (Side Side Side) condition. For example, if you have triangles with sides 4

cm, 6 cm, and 8 cm, it doesn't matter what order you draw them—there's only one way to arrange these side lengths for a triangle, so all triangles with those three lengths are congruent.

You don't always need to measure all three sides. Sometimes, two sides and the angle between them—called SAS (Side Angle Side)—are enough. If two triangles each have two sides of 6 cm and 5 cm and the included angle of 30° , there is only one possible size and shape for such a triangle.

But if the angle you know is not between those two sides (SSA condition), it may not be enough. You could sometimes draw two different, non-congruent triangles using the same two side lengths and an angle that is not included between those sides.

Other Conditions for Triangle Congruence

There are other ways to check for triangle congruence:

- **SAS (Side Angle Side):** Two sides and the included angle are equal.
- **ASA (Angle Side Angle):** Two angles and the included side are equal.
- **AAS (Angle Angle Side):** Two angles and a side (not between the angles) are equal.
- **RHS (Right angle, Hypotenuse, Side):** For right-angled triangles, if hypotenuse and one side are equal.

Measuring just all three angles (AAA) is not enough to guarantee congruence, because triangles with the same angle measures can have different sizes—they are similar, not congruent.

How is Congruence Expressed?

When triangles are congruent, we write $\triangle ABC \cong \triangle XYZ$. The order matters: the first letter matches the first, the second to the second, and the third to the third, showing which vertices, sides, and angles correspond with each other.

For rectangles or any quadrilaterals, the idea is the same: the corresponding sides and angles must be equal for congruence. Papers or shapes can always be rotated, flipped, or turned upside down before making this comparison.

Isosceles and Equilateral Triangles

In an isosceles triangle, two sides are equal. The angles opposite to those equal sides are also equal. For example, in triangle ABC, if $AB = AC$ and $\angle A = 80^\circ$, then angles B and C are equal, and you can find their value since the sum of angles in a triangle is always 180° .

Equilateral triangles have all three sides the same and all three angles are equal as well. Every angle in an equilateral triangle is always 60° . You can use the property that angles opposite equal sides are equal, and since the total sum must be 180° , dividing by three gives 60° for each angle.

Summary of Important Points

- Congruent figures have exactly the same shape and size and can overlap perfectly when superimposed.
- SSS, SAS, ASA, AAS, and RHS are all valid criteria for triangle congruence.
- SSA and AAA are not sufficient to prove congruence.
- Angles opposite to equal sides in a triangle are equal.
- The side opposite the right angle in a right triangle is called the hypotenuse.
- All angles in an equilateral triangle are 60° .

Congruence is not just for math problems. You can see congruent triangles in real life, such as in bridge designs, rangoli patterns, museum structures like the Louvre pyramid, the Pyramids of Giza, or the Howrah Bridge. Their sturdy and balanced designs often use congruent triangles for symmetry and strength.

Some problems also ask you to identify corresponding sides or angles, to state the congruence clearly, or to solve for missing angles using the properties above. Practising such exercises helps you spot congruence faster and apply it to a variety of geometric shapes.

Knowing how to spot and prove congruence is a valuable skill for all geometry and for mathematical reasoning in general.